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Edge-Based Energy-Efficient IoT Smart Waste Management System

Progress Presentation 1 · 50% Milestone

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GAP FOCUS

Joint Adaptive Control Policy

HARDWARE

ESP32 + 4 Sensors (Working Prototype)

PIPELINE

IoT → MQTT → Edge → Live Dashboard

PROBLEM STATEMENT & RESEARCH GAP

CURRENT IMPLEMENTATIONS FAIL

Existing IoT approach (Thakker & Shaikh 2015; Shyam et al. 2017):

Bin Node → transmit raw sensor data every 60s → Cloud → Dashboard

This creates three unsolved problems:

- Battery drain — Wi-Fi TX draws 240mA peak; fixed 60s rate = ~62 day battery on 2000mAh
- Bandwidth waste — stable bin at 30% fill still transmits 60×/hr; 95% of TXs carry no actionable data
- No adaptive intelligence — existing systems (Longhi 2012; Sankeerth 2020) don't adjust based on battery, fill velocity, or channel quality

RESEARCH GAP (GAP 1)

Closest prior art — Giordano et al. (ENSsys'23):

FSM + AIMD adapts sampling rate only (one output)
No deadband, urgency override, AoI floor, or bin context

Our contribution — joint 4-output policy on one node:

T*	TX interval — AIMD on battery SoC
ΔF*	Deadband — fill velocity adaptive
U	Urgency — fill + gas + rate signal
AoI	Max suppression floor per FSM state

HIGH-LEVEL SYSTEM ARCHITECTURE

Smart Bin Node	Edge Gateway	Cloud Backend	Web Dashboard	Mobile App
ESP32 + Sensors	Laptop (PP1) / Pi (PP2)	Node.js + SQLite	React + Vite	React Native Expo
► HC-SR04 fill level	► Rule-based filtering	► REST API	► Live FSM state	► Collection alerts
► MQ-135 / MQ-5 gas	► AIMD + deadband FSM	► Socket.io fanout	► TX comparison	► Issue reporting
► DHT22 temp/humidity	► Cluster arbitration	► Event storage	► Gas/fill charts	► Schedule view
► Battery monitoring	► Adaptive TX control	► Authentication	► Battery control	► (PP2 milestone)

PP1 scope: Smart Bin Node + Cloud Backend + Web Dashboard live. Edge Gateway (Raspberry Pi) + Mobile App deferred to PP2 per WBS.

JOINT ADAPTIVE CONTROL POLICY — OVERVIEW (4 OUTPUTS)

ALERT	**T* = 30s** ΔF* = 1%	*fill ≥ 80% OR gas > threshold OR dF/dt > 10%/hr*
ACTIVE	**T* = 90s** ΔF* = 3%	*fill ≥ 60% OR dF/dt > 2%/hr*
STABLE	**T* = 300–1800s (AIMD)** ΔF* = 1–10%	*Default state — AIMD governs interval from battery SoC*
CONSERVE	**T* = 600s** ΔF* = 5%	*battery < 30% (ALERT urgency override always wins)*

AIMD: batt > 80% → T* ÷ 2 | batt < 70% → T* + 60s | 70–80%: hold **Deadband:** ΔF* = max(1%, 10% − dF/dt × 0.5) **AoI floor enforced per state**

OUTPUT 1: T* — TRANSMISSION INTERVAL (AIMD Control Law)

What it is

T* is how long the ESP32 waits before the next transmission. It adapts based on battery level using AIMD — the same control law TCP uses for congestion.

Battery = congestion signal:
batt > 80% → halve T* (send more, you have power)
batt < 70% → add 60s (send less, conserve)
70–80% → hold (hysteresis, stop thrashing)

Walk-through example

Start:	T* = 300s, battery = 90%
90% > 80%:	300 × 0.5 = 150 → floored to 300s
battery → 65%:	300 + 60 = 360s
still 65%:	360 + 60 = 420s (climbing toward 1800s cap)
battery → 85%:	420 × 0.5 = 210 → floored to 300s

In config.h

firmware/include/config.h

```
#define AIMD_MIN   300000UL  // floor: 300s
#define AIMD_MAX  1800000UL  // ceiling: 1800s
#define AIMD_AI    60000UL   // additive +60s
#define AIMD_MD        0.5f  // multiplicative ×0.5
```

In fsm.h — STABLE branch

firmware/src/fsm.h (STABLE else-branch)

```
if (ctx.state != State::STABLE)
    ctx.txInterval = AIMD_MIN; // reset on re-entry
if (battPct > BATT_AIMD_MD_THRESH) {  // >80%
    ctx.txInterval *= AIMD_MD;         // halve
    ctx.txInterval = max(txInterval, AIMD_MIN);
} else if (battPct < BATT_AIMD_AI_THRESH) {//70%
    ctx.txInterval = min(txInterval+AIMD_AI, AIMD_MAX);
```

In main.cpp — transmission gate:

firmware/src/main.cpp

```
bool timerFired =
    (now - lastTxMs >= fsm.txInterval);
```

OUTPUT 2: ΔF* — DYNAMIC DEADBAND (Send-on-Delta, Miskowicz 2006)

What it is

Even when the timer fires, we don't transmit if fill hasn't actually changed enough. ΔF* is the minimum fill change needed to trigger a TX.

When the bin is stable → wide deadband (suppress HC-SR04 noise).
When filling fast → narrow deadband (catch real changes).

Formula: ΔF* = max(1%, 10% − dF/dt × 0.5)

dF/dt	ΔF*	Effect
0 %/hr	10%	Bin stable — suppresses ±1cm sensor noise
4 %/hr	8%	Slow fill — still ignores small fluctuations
10 %/hr	5%	Medium fill — catches most real changes
18 %/hr	1%	Fast fill — transmits on almost every reading

In fsm.h — computeDeadband()

firmware/src/fsm.h

```
static float computeDeadband(float dFdt) {
  // dFdt is in %/s — convert to %/hr first
  float dFdt_hr = dFdt * 3600.0f;
  float db = 10.0f - dFdt_hr * 0.5f;
  return max(1.0f, db); // never fully suppress
```

dF/dt calculation — from fsm.h:

firmware/src/fsm.h (top of fsmUpdate)

```
uint32_t dt = nowMs - ctx.lastSampleMs;
// fill velocity in %/s
float dFdt = (fillPct - ctx.prevFill) / (dt/1000.0f);
deadbandOut = computeDeadband(dFdt);
```

Used in main.cpp — transmission gate:

firmware/src/main.cpp

```
// how much has fill moved since last TX?
float fillDelta = fabs(r.fillPct - lastTxFill);
bool deadbandCrossed = (fillDelta >= deadband);
// TX fires if timer OR deadband OR urgency
```

OUTPUT 3: URGENCY OVERRIDE (Priority Always Beats Energy Conservation)

Priority evaluation order — runs every loop tick:

1	Urgency check	fill ≥ 80% OR gas > threshold OR dF/dt > 10%/hr → ALERT
2	Battery check	battery < 30% → CONSERVE (only if urgency false)
3	Activity check	fill ≥ 60% OR dF/dt > 2%/hr → ACTIVE
4	Default	none of above → STABLE (AIMD runs here)

The stateChanged guard — critical detail:

Without stateChanged, every loop tick while fill ≥ 80% would transmit (~55 times/min — worse than baseline).

With stateChanged: urgency fires ONCE on ALERT entry → one immediate TX. After that, the 60s AoI floor (txInterval) controls cadence.

Result: fast alert + controlled energy.

In fsm.h — urgency check (ALWAYS first):

firmware/src/fsm.h (fsmUpdate — priority order)

```
// 1. URGENCY — runs before battery check
if (fillPct >= FILL_ALERT         // ≥80%
    || gasRaw135 > GAS135_THRESHOLD
    || gasRaw5   > GAS5_THRESHOLD
    || (dFdt*3600.0f) > DFDT_ALERT) {//10%/hr
    ctx.state = State::ALERT;
```

In main.cpp — stateChanged gate:

firmware/src/main.cpp (ADAPTIVE branch)

```
static State prevState = State::STABLE;
bool stateChanged = (fsm.state != prevState);
prevState = fsm.state;
// fire ONCE on entry, not every tick
if ((urgent && stateChanged) ||
    timerFired || deadbandCrossed) { publish(); }
```

Reason strings (visible in dashboard):

firmware/src/main.cpp

```
reason = fillPct>=FILL_ALERT  ? "urgency-fill"
       : gasRaw135>THRESH135  ? "urgency-gas135"
       : gasRaw5>THRESH5      ? "urgency-gas5"   : "urgency-
```

OUTPUT 4: AoI FLOOR (Maximum Suppression — Node Can Never Go Silent)

What it is

AoI = Age of Information — how stale is the data at the receiver?

Without an AoI floor, a bin that is 75% full but not changing could go silent indefinitely — deadband never crossed, no urgency. The dashboard would not know if the node is alive or dead.

The AoI floor is a forced heartbeat: transmit at least once every N seconds, regardless of deadband.

AoI floor values per state:

ALERT	60s	Overflow risk — operator MUST have current data
ACTIVE	5 min	Filling trend trackable for overflow prediction
STABLE	30 min	Heartbeat only — confirms node is alive
CONSERVE	15 min	Battery critical — less frequent but never silent

In config.h — AoI constants:

firmware/include/config.h

```
#define AoI_ALERT     60000UL  //  60s
#define AoI_ACTIVE   300000UL  //   5min
#define AoI_STABLE  1800000UL  //  30min
#define AoI_CONSERVE 900000UL  //  15min
```

In fsm.h — AoI IS the txInterval per state:

firmware/src/fsm.h (state transitions)

```
if (fill >= FILL_ALERT || gas || rate) {
    ctx.txInterval = AoI_ALERT;  // 60s floor
} else if (batt <= BATT_LOW) {
    ctx.txInterval = AoI_CONSERVE; // 15min floor
} else if (fill >= FILL_ACTIVE) {
    ctx.txInterval = AoI_ACTIVE;  // 5min floor
```

In main.cpp — enforced via timer:

firmware/src/main.cpp

```
// AoI floor = txInterval in non-STABLE states
bool timerFired = (now-lastTxMs >= fsm.txInterval);
```

AoI insight: In ALERT/ACTIVE/CONSERVE, txInterval IS the AoI floor. In STABLE, AIMD governs txInterval but AIMD_MAX (1800s) acts as the implicit AoI ceiling.

PROOF OF CONCEPT — CURRENT IMPLEMENTATION

ESP32 Serial Monitor — FSM State Transitions

```
[WiFi] connecting to Redmi Note 14...
[WiFi] connected, IP=10.97.85.244
[RTC] fresh boot, initializing
[MQTT] connecting... ok
[MQ] warmup complete
[FSM] state=STABLE interval=300s
[TX] {"state":"STABLE","fill":12.4,"reason":"timer"}
[FSM] state=ACTIVE interval=90s
[TX] {"state":"ACTIVE","fill":64.2,"reason":"timer"}
[FSM] state=ALERT interval=60s
[TX] {"state":"ALERT","fill":82.1,"reason":"urgency-fill"}
[FSM] CONSERVE (batt=25%)
```

MQTT PIPELINE — VERIFIED

ESP32	Sensors read + FSM evaluated every 1s
MQTT	Mosquitto broker @ 10.97.85.244:1883
Node	Ingests + stores 15-field JSON to SQLite
Dash	Socket.io → live update every transmission

bin/bin01/data — Live MQTT Payload


```
{"node_id":"bin01","mode":"ADAPTIVE",
 "state":"STABLE","fill_pct":23.4,
 "gas_135":1280,"gas_5":2340,
 "batt_pct":80,"temp_c":29.8,
 "interval":300,"deadband":9.8,
 "dfdt":0.12,"reason":"timer","tx_count":7}
```


✓ WiFi + MQTT verified ✓ FSM state transitions live ✓ Dashboard real-time ✓ Battery slider controls CONSERVE ✓ 4 sensors operational

TECHNOLOGY STACK & STANDARDS APPLIED

IoT Firmware Layer

- ESP32-D0WD-V3 NodeMCU 38-pin — 240MHz dual-core, RTC memory for FSM state persistence
- HC-SR04 fill level (median-of-3 filter) · MQ-135 VOC · MQ-5 LPG/methane · DHT22 temp/humidity
- PlatformIO + Arduino framework · PubSubClient (MQTT) · ArduinoJson · Espressif ESP-IDF deep sleep API

Communication & Backend

- MQTT protocol (ISO/IEC 20922) via Mosquitto broker — lightweight pub/sub, QoS 0, 200-byte payloads
- Node.js + Express RESTful API (REST architectural style, HTTP/1.1) · Socket.io WebSocket fanout
- better-sqlite3 — structured event store; 15-field normalised schema per reading

Standards & Best Practices

- IEEE citation format — 12 peer-reviewed references (ACM, IEEE IoT Journal, PLOS ONE, ENSsys)
- FSM design pattern (Giordano ENSsys'23) · AIMD control law (TCP Reno) · Send-on-delta (Miskowicz 2006)
- Code: modular headers (config.h, fsm.h, sensors.h) · inline comments · separation of concerns · version-pinned libs

REQUIREMENTS • WBS PROGRESS • RISK MITIGATION

ID	Requirement	Status
FR1	IoT node captures fill, gas, temp, battery	✓ Done
FR2	Node sends via MQTT lightweight protocol	✓ Done
FR3	Edge gateway aggregates multiple bins	PP2
FR4	Edge applies rule-based FSM filtering	✓ Laptop
FR5	Edge adapts TX using battery + network	✓ Done
FR6	Web app displays bin status + trends	✓ Done
FR7	Web app supports route-planning	PP2
FR8	Mobile app — schedules, alerts, reporting	PP2
FR9	Backend stores data securely	✓ Done

WBS at 50% milestone:



RISK MITIGATION

Sensor inaccuracy / MQ drift	Med
Gateway failure	Med
Network instability	Med
Scope creep (mobile/cluster deferred)	Low
Boot-mode conflict (strapping pins)	Low

COMMERCIALIZATION & NEXT MILESTONE

COMMERCIALIZATION

Target Markets

- Municipal councils & private waste firms
- Residential complexes, universities, hospitals

Revenue Model

- One-time hardware: smart-bin + gateway cluster
- Monthly SaaS: dashboard + analytics + support
- Managed services + municipal integrations

Value Proposition

- ~60% TX reduction → lower data costs + longer battery
- Edge resilience — works without internet connectivity

PP2 MILESTONE TARGETS

WBS-4 — Raspberry Pi edge gateway — rule engine, cluster arbitration, OVERRIDE_T feedback loop

WBS-7 — Mobile app — collection schedules, push alerts, issue reporting (React Native)

WBS-8 — System integration — 3-node cluster simulation, end-to-end evaluation

WBS-9 — Quantitative evaluation — measured TCR, AR, BLEF against fixed-interval baseline

SDG Alignment: SDG 11 (Sustainable Cities) + SDG 12 (Responsible Consumption). **Key differentiator:** First joint AIMD + deadband + urgency + AoI policy on a single ESP32 node applied to smart waste.